



Research Paper

Recycling excavated soil waste into low-carbon alkali-activated materials

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ABSTRACT

Rapid urbanization generates billions of tons of excavated soil waste annually from underground developments, presenting significant sustainability challenges. The objective of this study was to develop a sustainable recycling method for soil waste. Soil waste was calcined to transform its kaolin content into metakaolin. Then, the calcined soil, supplemented with ground granulated blast-furnace slag and uncalcined soil, was used to produce low-carbon alkali-activated construction materials. An optimal formulation of the alkali-activated materials was identified through strength, efflorescence, and microstructure testing, alongside preliminary carbon footprint and economic evaluations. Results showed that alkali-activated calcined soil alone exhibited middle-class strength and a high efflorescence risk. Adding slag altered the alkali-activated products from sodium aluminosilicate hydrate to a coexistence of sodium aluminosilicate hydrate and calcium aluminosilicate hydrate, reducing porosity and refining pore size due to the increased chemically bound water, thus enhancing strength and efflorescence resistance, as well as reducing CO₂ emissions and cost per MPa. When slag was added, substituting 25 % calcined soil with uncalcined soil increased the calcium aluminosilicate hydrate content, enhancing sustainability without sacrificing performance, despite the inert nature of uncalcined soil. The binder with the optimized formulation, i.e. 52.5 % calcined soil, 30 % slag, and 17.5 % uncalcined soil, may be an excellent alternative to the conventional cement-based binder, concerning both mechanical performance (high early compressive strength of 40 MPa after one day) and carbon footprint (42 % reduction in CO₂ emissions compared to the blended binder combining 70 % ordinary Portland cement with 30 % fly ash). These advantages show that this method appears to be a promising option for recycling excavated soil waste and reducing CO₂ emissions.

1. Introduction

With rapid urbanization, massive volumes of soil waste are excavated during underground space development such as the construction of underground buildings and subways. It has become the main component of current construction and demolition waste (CDW), posing a significant sustainability challenge due to the lack of large-scale recycling methods. Annually, China produces 2–3 billion tons of excavated soil waste, over half of CDW (Wu et al., 2022; Xiao et al., 2021). The European Union generates 460 million tons per year of this waste stream, constituting 54 % of CDW (Becker et al., 2020; Villoria Sáez and Osmani, 2019). Over 95 % of excavated soil waste ends up in dumping sites, resulting in high transportation costs, significant land resource occupation, pollution, and safety hazards (Zhang et al., 2020; Zhou et al., 2021; Duan et al., 2018). In 2015, a landslide from excessive soil waste dumping in Shenzhen caused 33 buildings to collapse and 77 deaths (Gao et al., 2018; Yang et al., 2017). Large-scale recycling of soil

waste is crucial to address these issues.

Recycling CDW as construction materials, for example as recycled aggregates, holds significant potential for large-scale use (Martínez-Lage et al., 2020; Pu et al., 2023; Puthusseray et al., 2017). However, for excavated soil waste, a notable challenge lies in developing construction materials that are characterized by both adequate strength and high soil content, which limits its large-scale recycling. Current studies report two primary pathways for recycling soil waste into construction materials, each presenting its own limitations. One path uses uncalcined soil waste as an inert filler for low-strength construction materials (Gao et al., 2023; Hu et al., 2022; Qian et al., 2019; Santos Nascimento et al., 2021; Shen et al., 2024). The limited strength (~10 MPa) restricts broader use. Another path uses calcined soil as a supplementary cementitious material, partially replacing Portland cement in concrete production (Borno et al., 2024; Dhandapani et al., 2023; Priyadharshini et al., 2018; Zhou et al., 2021, 2023b, 2024a). Soil waste is rich in clay (Hu et al., 2023), and calcined clay can offer glassy Si and Al which react chemically with

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the calcium hydroxide produced during cement hydration. This reaction not only reduces calcium hydroxide, which is a vulnerability for the final concrete's microstructure, but also forms additional calcium silicate hydrate (C-S-H) and calcium aluminosilicate hydrate (C-A-S-H), the primary binding phases in concrete. Although calcination consumes energy and emits CO₂, calcined soil's CO₂ emissions are still much lower than Portland cement's (Zhou et al., 2021). However, through this recycling method only up to 30 % of cement can be replaced by calcined soil. Exceeding this ratio significantly compromises the mortar or concrete's strength (Borno et al., 2024; Zhou et al., 2021).

Alkali activation is one potential method to recycle excavated soil waste into robust construction materials with high soil content. Alkali-activated materials (AAMs), produced through the reaction of glassy Si and Al in calcined clay with a small amount of alkali-activator, have gained worldwide attention as a low-carbon alternative to cement (Jiang et al., 2024; Khalifa et al., 2020). Mining and dredging wastes containing clay minerals have been successfully used to produce AAMs (Longhi et al., 2022b; Pradhan et al., 2023; Žibret et al., 2023). Notably, global construction consumes 4.1 billion tons of cement annually, contributing to 8 % of man-made CO₂ emissions (Coffetti et al., 2022; Monteiro et al., 2017). The significant amounts generated and the clay-rich composition of excavated soil waste make it a potential ingredient for AAMs manufacturing in substitution of cement.

Besides the alkali activation of calcined clay alone, there is much potential for developing alkali-activated systems that incorporate both clay and a calcium source such as ground granulated blast-furnace slag (GGBS), lime, or even cement (Alventosa et al., 2022; Khalifa et al., 2020; Li et al., 2023; Matalkah et al., 2021; Peng et al., 2022; Shilar et al., 2023). Alkali activation of calcined clay alone results in a low-calcium system (Na₂O-Al₂O₃-SiO₂ system) and may exhibit drawbacks such as relatively low strength and long setting times at ambient temperatures and a high efflorescence risk. Efflorescence resulting from free alkali migrating to the surface and reacting with CO₂ to form carbonates (Na₂CO₃) significantly challenges AAM durability, not only affecting the appearance but also potentially causing cracking and deterioration due to the crystallization and expansion of Na₂CO₃ within the subsurface (Ge et al., 2023; Longhi et al., 2022a; Zhang et al., 2018). Incorporating 10–30 % of a CaO-rich precursor to obtain a CaO-Na₂O-Al₂O₃-SiO₂ system improves setting time, strength, and efflorescence resistance. This type of AAM was successfully demonstrated in aircraft concrete pavement construction at Wellcamp Airport in Australia (Glasby et al., 2015). Soil waste lacks calcium; therefore, using calcined soil waste supplemented by GGBS as a Ca silicate source may enhance the performance of concrete products compared to using calcined soil waste alone.

Furthermore, to reduce energy consumption, carbon emissions, and costs, it is worth exploring replacing calcined soil waste with uncalcined soil waste. Although uncalcined soil waste is considered inert, its appropriate incorporation may not cause performance loss. Existing studies have shown that fillers such as limestone powder, basalt powder, waste marble powder, and recycled aggregate powder can replace 25–50 % of the calcined clay in alkali-activated calcined clay without compromising its properties, although these fillers showed much lower reactivity compared to calcined clay (Kabirova et al., 2023; Liu et al., 2023; Miyan et al., 2024; Omur et al., 2024; Perez-Cortes and Escalante-Garcia, 2020).

For large-scale recycling of excavated soil waste and possibly reducing the construction sector's reliance on carbon-intensive cement, this study proposes recycling excavated soil waste into low-carbon AAMs. To achieve this goal, the soil waste was calcined and alkali-activated, supplemented by GGBS for calcium addition and thus enhanced performances, and partially replaced by uncalcined soil for enhanced sustainability. The effects of the addition of different amounts of GGBS and uncalcined soil on strength, efflorescence resistance, microstructure, CO₂ emissions, and economic costs were investigated. The optimized formulation was identified by evaluating multiple

performance criteria. Additionally, the carbon footprints and economic costs of the AAMs were compared to those of conventional cement binders.

2. Materials and methods

2.1. Materials

The soil waste obtained from the excavation of subway stations in Shenzhen was ground to below 0.3 mm and then calcined at 800 °C for 2 h to produce calcined soil. Table S1 in the Supplementary Material shows the chemical composition of the soil before and after calcination determined by X-ray fluorescence (XRF) using a Panalytical Axios XRF spectrometer. Some impurities and chemically bound water in the uncalcined soil can be removed by calcination, so the SiO₂ and Al₂O₃ contents increased after calcination. Fig. S1 in the Supplementary Material shows the X-ray Diffraction (XRD) results of the soil before and after calcination determined using a Rigaku Ultima IV X-ray diffractometer. Using the Rietveld method (Yan et al., 2018), XRD analysis revealed that uncalcined soil primarily consisted of 42.1 % quartz (SiO₂) and 57.9 % kaolinite (Al₂Si₂O₅(OH)₄), whereas the calcined soil contained 47.1 % quartz and 52.9 % amorphous phases, determined using zinc oxide (ZnO) as an internal standard with a ZnO-to-soil mass ratio of 1:4. Calcination led to a slight increase in the quartz content, while kaolinite (Al₂Si₂O₅(OH)₄) transformed into metakaolin (Al₂O₃·2SiO₂), an amorphous phase undetectable by XRD. The loss of chemically bound water resulted in a slight reduction in the amorphous content compared to the original kaolinite content. Also, the reactive SiO₂ and Al₂O₃ contents of the calcined soil, determined by the NaOH dissolution method (Chen-Tan et al., 2009; Kuenzel and Ranjbar, 2019; Luan et al., 2024a), were 27.3 % and 23.1 %, respectively, consistent with the amorphous content. Table S1 also provides the oxide composition of the GGBS employed in the tests, analysed by XRF.

The particle size distribution of the excavated soil waste, determined using a Linkoptik LT2200 Laser Particle Analyzer, is shown in Fig. S2 in the Supplementary Material. The particle size distributions of both uncalcined and calcined soils were similar and closely resembled those of cement reported in previous studies (Koroglu et al., 2021; Ma et al., 2023; Wu et al., 2022; Zhou et al., 2023a).

The alkali-activated solution employed in the tests was a mixture of a sodium silicate solution (Na₂O = 13.75 wt%; SiO₂ = 29.99 wt%; H₂O = 56.26 wt%) with a sodium hydroxide (NaOH) solution (NaOH = 17.2 wt %) and was prepared 24 h in advance before use. The mass ratio of sodium silicate solution to NaOH solution was set at 1.3. For all mixtures, the Na₂O to precursor mass ratio was set at 7 %, and the activator modulus (mass ratio of SiO₂ to Na₂O) was set at 1.25. For mortar preparation, the particle size of the aggregate (river sand) was smaller than 2.5 mm. For concrete preparation, the particle sizes of the aggregates (river sand and granite crushed stone) were distributed as follows: 0–5 mm (50 %), 5–10 mm (30 %), and 10–20 mm (20 %), respectively.

2.2. Mix proportion and sample preparation

For all mixtures, the mass ratio of alkali-activated solution to precursor (i.e. soil and GGBS) was set at 0.516, which led to a water to solid mass ratio of 0.3, where the solid included the precursor, sodium silicate and NaOH.

The influence of the GGBS content on the performance of calcined soil waste-based AAMs was assessed in this study. Four levels (0 %, 10 %, 20 %, and 30 % by mass) of GGBS content were used. These trial groups were labeled as C100G0, C90G10, C80G20, and C70G30, respectively. For example, C70G30 meant that the precursor consists of 70 % calcined soil and 30 % GGBS. It should be noted that preliminary experiments showed that the mixture experienced flash setting when the GGBS content exceeded 40 %, so the GGBS content was limited to a maximum of 30 %. Mortar samples were used for flowability, setting

time, flexural and compressive strength, and scanning electron microscope (SEM) tests. In mortar samples, the sand to precursor ratio was set at 1.5. Paste samples were used for efflorescence, water absorption, Fourier transform infrared spectrum (FTIR), thermogravimetric analysis (TGA), and mercury intrusion porosimetry (MIP) tests. They were prepared in a planetary laboratory mixer using the following procedure: initially, the precursors (soil and GGBS) and the alkali-activated solution were mixed at a slow speed for 30 s. Next, sand was added (excluding paste mixtures), and the slow-speed mixing continued for an additional 30 s. Then, the mixture attached to the rotating blade and the mixer wall was scraped back into the mixer, and high-speed mixing was performed for 60 s. Later, the fresh mixtures were poured into molds, vibrated for 60 s (20 s for paste mixtures), and sealed using plastic films. After 24 h, specimens ($40 \times 40 \times 160 \text{ mm}^3$ prisms for mortar and cylinders with a diameter of 20/28 mm and a height of 35/55 mm for paste) were demolded and cured under standard curing conditions (temperature = 20°C ; relative humidity > 95 %) until testing.

The above mentioned experiment for assessing the influence of GGBS content showed that the blend of 70 % calcined soil and 30 % GGBS exhibited the best performance, so it was used as the reference mixture to further assess the influence of uncalcined soil waste content. The GGBS content was kept at 30 % by mass, and the uncalcined soil replaced calcined soil partially or fully. Five levels of uncalcined soil content (0 %, 17.5 %, 35 %, 52.5 %, and 70 % by mass) were considered, resulting in corresponding calcined soil contents of 70 %, 52.5 %, 35 %, 17.5 %, and 0 %, respectively. These trial groups were labeled as C70G30U0, C52.5G30U17.5, C35G30U35, C17.5G30U52.5, and C0G30U70, respectively. For example, C17.5G30U52.5 meant that the precursor consisted of 17.5 % calcined soil, 30 % GGBS, and 52.5 % uncalcined soil. Concrete samples were prepared for all groups to undergo slump, compressive strength, and SEM tests. In the concrete samples, the aggregate to paste ratio was 7:3. The concrete samples were prepared in a compulsion-type concrete mixer. First, soil and GGBS were mixed for 30 s. Then, the alkali-activated solution was added, and the mixing continued for an additional 30 s. Next, the aggregate was added, and the mixing continued for an additional 150 s. The fresh mixtures were poured into the 100 mm cube molds, vibrated for 60 s, and then sealed using plastic films. After 24 h, all specimens were demolded and cured under standard curing conditions until testing. Paste samples, excluding aggregates, were prepared for selected groups.

2.3. Test methods

The experimental tests included: flowability, setting time, flexural and compressive strength, and SEM tests for mortar samples; efflorescence, water absorption, FTIR, TGA and MIP tests for paste samples; and slump, compressive strength, and SEM tests for concrete samples.

The flowability of the fresh mortar was assessed using the flow table test, following the Chinese standard GB/T 2419-2005, similar to the European standard EN 1015-3:2007. The process involved pouring the fresh mortar into a mini conical mold placed on a flow table, lifting the mold vertically, and then jolting the flow table 25 times to spread out the mortar. The mean value of two perpendicular diameters was reported as the flow value. The setting time of the fresh mortar was measured using the penetration rod test, following the Chinese standard JGJ/T 70-2009, which is equivalent to the European standard EN 1015-9:2007. The setting time was reported as the time when the mortar reaches a penetration resistance of 0.5 MPa.

The one-day and 28-day strength of the hardened mortar, including flexural and compressive strength, were determined following the Chinese standard GB/T 17671-2021, equivalent to the European standard EN 196-1:2016. Three-point flexural testing on $40 \times 40 \times 160 \text{ mm}^3$ prism samples was used. The loading rate was 50 N/s and the span was 100 mm. The flexural strength was averaged from three replicated specimens. The compressive strength was determined using the remaining six prism halves with a platen area of $40 \times 40 \text{ mm}^2$ and a

loading rate of 2400 N/s. The mortar fragments were analyzed using a Phenom ProX scanning electron microscope after drying and gold spraying.

Efflorescence tests were conducted following the methods of Longhi et al. (2021). The efflorescence resistance was shown by the visual efflorescence formation and the efflorescence potential assessed through leaching tests. The efflorescence formation test was conducted on cylinder paste specimens with a diameter of 28 mm and a height of 55 mm in a chamber with an ambient temperature of 20°C and relative humidity of 60 %. After curing for 28 days, these specimens were submerged in water, and the water level was maintained at 5 mm by adding water twice daily to compensate for water absorption and evaporation. The humidity gradient can in fact accelerate efflorescence formation. The efflorescence potential test was carried out on cylinder paste specimens with a diameter of 20 mm and a height of 35 mm. After curing for 28 days, these specimens were individually and fully immersed in 500 ml of deionized water. After 28 days, the sodium (Na) concentration of the leaching solution was determined by Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES) analysis using a Thermo Scientific iCAP 7400 instrument. A lower Na concentration indicates a lower efflorescence potential.

Water absorption was measured on cylinder paste specimens with a diameter of 28 mm and a height of 55 mm, following the method of Longhi et al. (2021). After 28 days of curing, the specimens were dried at 105°C for 24 h. Then, they were submerged in water, and the water level was maintained at 5 mm. The high capillary pressure pushed the water diffusion from the bottom to the top, filling pores. The variations in absorbed water amount over time were reported, as well as the open porosity, i.e. the proportion of the final amount of water absorbed to the volume of the sample.

FTIR can identify specific chemical species in AAMs according to the vibration band around 1000 cm^{-1} , which corresponds to (Si/Al)–O–Si structures (Garcia-Lodeiro et al., 2011). TGA can determine the chemically bound water content, i.e., the mass loss between 105°C and 1000°C (Baert et al., 2008; Pane and Hansen, 2005). To prepare the samples for FTIR and TGA, the paste specimen, after 28 days of curing, was cut into pieces, and ground into powders with sizes less than $45 \mu\text{m}$. Subsequently, the powders were dried at 105°C for 24 h. FTIR analysis involved mixing 1 mg of the dried powder with 100 mg of KBr powder, pressing the mixed powder to obtain a tablet, and finally testing the tablet employing a Tianjin Gangdong FTIR-650S spectrometer to obtain the spectrum with wavenumbers between 400 and 4000 cm^{-1} and a resolution of 4 cm^{-1} . TGA was performed using 10 mg of the dried powder on a Beijing Hengjiu HCT-1 thermal analyzer, with a heating rate of $10^\circ\text{C}/\text{min}$ gradually increasing from ambient temperature to 1000°C .

The pore size distribution and MIP porosity of selected paste samples were assessed using an AutoPore Iv 9510 MIP analyzer, with pressures up to 421 MPa (61000 psi). A contact angle of 130° was assumed in this study. The samples were cut to approximately $10 \times 10 \times 10 \text{ mm}^3$ and then dried at 80°C for 48 h before testing.

The slump of the fresh concrete was measured using a slump cone, following the Chinese standard GB/T 50080-2016, which is equivalent to the European standard EN 12350-2:2019. The compressive strength of the hardened concrete was determined following the Chinese standard GB/T 50081-2019, which is equivalent to the European standard EN 12390-3:2009. Using a 2000-kN MTS testing machine, cubes with a size of 100 mm were tested with a loading rate of 0.5 MPa/s on days 1 and 28. The compressive strength value was averaged from three replicated specimens. The concrete fragments were analyzed using a Phenom ProX scanning electron microscope after drying and gold spraying.

2.4. Assessment of carbon emissions and costs

This study conducted a preliminary assessment of the CO_2 emissions

and costs associated with utilizing excavated soil waste to produce alkali-activated concrete. A cradle-to-gate life cycle perspective was employed, as illustrated in Fig. S3 in the Supplementary Material (Shobeiri et al., 2021; Turk et al., 2015; Van den Heede and De Belie, 2012). The functional unit was defined as 1 m³ of concrete. The CO₂ emissions and costs of the developed alkali-activated concrete were compared with those of the commonly used blended cement concrete in which 30 % of the cement was replaced with fly ash. This blended cement concrete offers reduced CO₂ emissions and cost compared to cement concrete using 100 % cement. Therefore, using 100 % cement concrete as the reference would overestimate the sustainability advantages of the developed alkali-activated concrete, while using blended cement concrete as the reference enables a more accurate evaluation of the sustainability advantages of the developed alkali-activated concrete. The compressive strength of the concrete produced in this study reached a maximum of 58 MPa, so the reference cement concrete was assumed to have a similar strength grade of 55 MPa, in order to assume that the two products would be used for the same types of applications.

Table 1 shows the data used for CO₂ emissions and cost calculation, including the unit prices, CO₂ emission factors, and transport distances of raw materials, along with the unit price and CO₂ emission factor of transport. The source of each data item is also provided in Table 1. The CO₂ emission factors and unit prices of NaOH, GGBS, calcined soil, uncalcined soil, cement, fly ash, aggregate, and transportation, as well as the unit price of sodium silicate solution, refer to the Chinese context. The CO₂ emission factor of sodium silicate was based on the global data from the Ecoinvent database reported in Habert and Ouellet-Plamondon (2016). A detailed literature survey (Abbas et al., 2020; Alsaman et al., 2021; Davidovits, 2015; Gao et al., 2017; Habert and Ouellet-Plamondon, 2016; Habert et al., 2011; Kanagaraj et al., 2022; Ling et al., 2024; Miyan et al., 2024; Perez-Cortes and Escalante-Garcia, 2020; Rodriguez-Morales et al., 2024; Shah et al., 2022; Teh et al., 2017; Turner and Collins, 2013) shows that despite numerous studies, the CO₂ emission factor of sodium silicate derives primarily from two sources: Fawer et al. (1999) and the Ecoinvent database. Fawer et al. (1999) reported a CO₂ emission factor of 1.066 kgCO₂/kg for European sodium silicate, while the Ecoinvent database gave a European emission

factor of 1.08 kgCO₂/kg and a global figure (excluding Europe) of 1.06 kgCO₂/kg, as seen in Habert and Ouellet-Plamondon (2016). Given the regional specificity of the data, the value of 1.06 kgCO₂/kg from the Ecoinvent database was used in this study. Shah et al. (2022) also applied this global figure as the CO₂ emission factor for sodium silicate in China. It is important to note that the value refers to anhydrous sodium silicate production, not that of the solution form. The reduced CO₂ emission factor from water addition must be considered (Alsaman et al., 2021; Davidovits, 2015; Habert and Ouellet-Plamondon, 2016). Ignoring this point would lead to an inappropriate CO₂ emission factor of sodium silicate solution (>1 kgCO₂/kg). This study employed a sodium silicate solution with Na₂O of 13.75 % and SiO₂ of 29.99 %. Consequently, its CO₂ emission factor was calculated as 472.4 kgCO₂/t, obtained by multiplying 1060 with the sum of Na₂O and SiO₂ percentages. The CO₂ emission factor for NaOH, specific to China, was sourced from Hong et al. (2014), with estimated values between 1.04 and 1.59 kgCO₂/t. The CO₂ emission factor of NaOH varies as a function of the electricity sources and production technologies. The average value of 1.32 kgCO₂/t was used in this study. The range of 1.04–1.59 kgCO₂/t is within the value range of 0.89–1.92 kgCO₂/t reported in other studies which are not specific for China (Alsaman et al., 2021; Font et al., 2020; Gao et al., 2017; Habert and Ouellet-Plamondon, 2016; Kanagaraj et al., 2022; Ling et al., 2024; Miyan et al., 2024; Perez-Cortes and Escalante-Garcia, 2020; Robayo-Salazar et al., 2018; Teh et al., 2017; Turner and Collins, 2013).

The calculations of the unit prices and CO₂ emission factors for the uncalcined soil and calcined soil are outlined in Table S2 in the Supplementary Materials. The CO₂ emissions and costs of calcined soil were calculated as the sum of transport, grinding, and calcination. As the soil was naturally sun-dried before grinding and the associated cost and CO₂ emissions are minimal, these were simplified to zero. According to Zhou et al. (2021), the transport distance from the construction site to the recycling plant was 15 km, grinding energy use was 347 MJ/t, and calcination energy use was 1844 MJ/t. Therefore, assuming grinding uses electricity and calcination employs coal, the calculated price and CO₂ emission factor of calcined soil were 16.3 USD/t and 230.2 kgCO₂/t, respectively. Also, the costs and emissions for uncalcined soil, incorporating only transportation and grinding, were estimated at 9.5 USD/t and 55.1 kgCO₂/t, respectively.

To meaningfully compare concrete products of different strengths, two factors E_f (CO₂ emissions per MPa, kgCO₂/m³/MPa) and C_f (cost per MPa, USD/m³/MPa), which are described in Eqs. (1) and (2), were utilized (Damineli et al., 2010; Qian et al., 2023). The CO₂ emissions per unit volume and cost per unit volume of concrete products with different strengths cannot be directly compared, as higher concrete strength results in a smaller cross-section of structural elements subjected to axial compressive loads and a reduced concrete volume. Therefore, the CO₂ emissions and cost incurred to deliver 1 MPa of compressive strength were employed as sustainability indicators to compare concrete products of different strengths. Lower values of E_f and C_f indicate a higher sustainability.

$$E_f = \frac{\text{CO}_2 \text{ emissions of } 1\text{m}^3 \text{ concrete}}{\text{Compressive strength}} \quad (1)$$

$$C_f = \frac{\text{Cost of } 1\text{m}^3 \text{ concrete}}{\text{Compressive strength}} \quad (2)$$

3. Results and discussion

3.1. Influence of the GGBS content on the performance of the mortar samples

The influence of the GGBS content on the 28-day compressive and flexural strengths of alkali-activated calcined soil-GGBS mortar is shown in Fig. 1a. Both compressive and flexural strengths consistently

Table 1
Calculation parameters for CO₂ emissions and costs.

	Unit price	CO ₂ emission factor	Transport distance
NaOH	400.0 USD/t (Assi et al., 2020)	1315.0 kgCO ₂ /t (Hong et al., 2014)	50.0 km (Zhang et al., 2017)
Sodium silicate solution	145.0 USD/t (Luan et al., 2024b)	472.4 kgCO ₂ /t (Habert and Ouellet-Plamondon, 2016)	50.0 km (Zhang et al., 2017)
GGBS	36.2 USD/t (Ccement, 2023)	132.3 kgCO ₂ /t (Chen et al., 2010)	50.0 km (Zhang et al., 2017)
Calcined soil	16.3 USD/t (Zhou et al., 2021)	230.2 kgCO ₂ /t (Zhou et al., 2021)	15.0 km (Zhou et al., 2021)
Uncalcined soil	9.5 USD/t (Zhou et al., 2021)	55.1 kgCO ₂ /t (Zhou et al., 2021)	15.0 km (Zhou et al., 2021)
P-O 42.5 cement	72.5 USD/t (Ccement, 2023)	732.0 kgCO ₂ /t (T/CBMF 27, 2018)	50.0 km (Zhang et al., 2017)
Fly ash	21.7 USD/t (Luan et al., 2024b)	218.0 kgCO ₂ /t (Chen et al., 2010)	50.0 km (Zhang et al., 2017)
Aggregate	8.7 USD/t (Luan et al., 2024b)	4.0 kgCO ₂ /t (T/CBMF 27, 2018)	50.0 km (Zhang et al., 2017)
Transport	1.2 × 10 ⁻⁴ USD/(kg·km) (HIGHWAY MINISTRY OF TRANSPORT, 2021)	1.4 × 10 ⁻⁴ kgCO ₂ /(kg·km) (T/CBMF 27, 2018)	

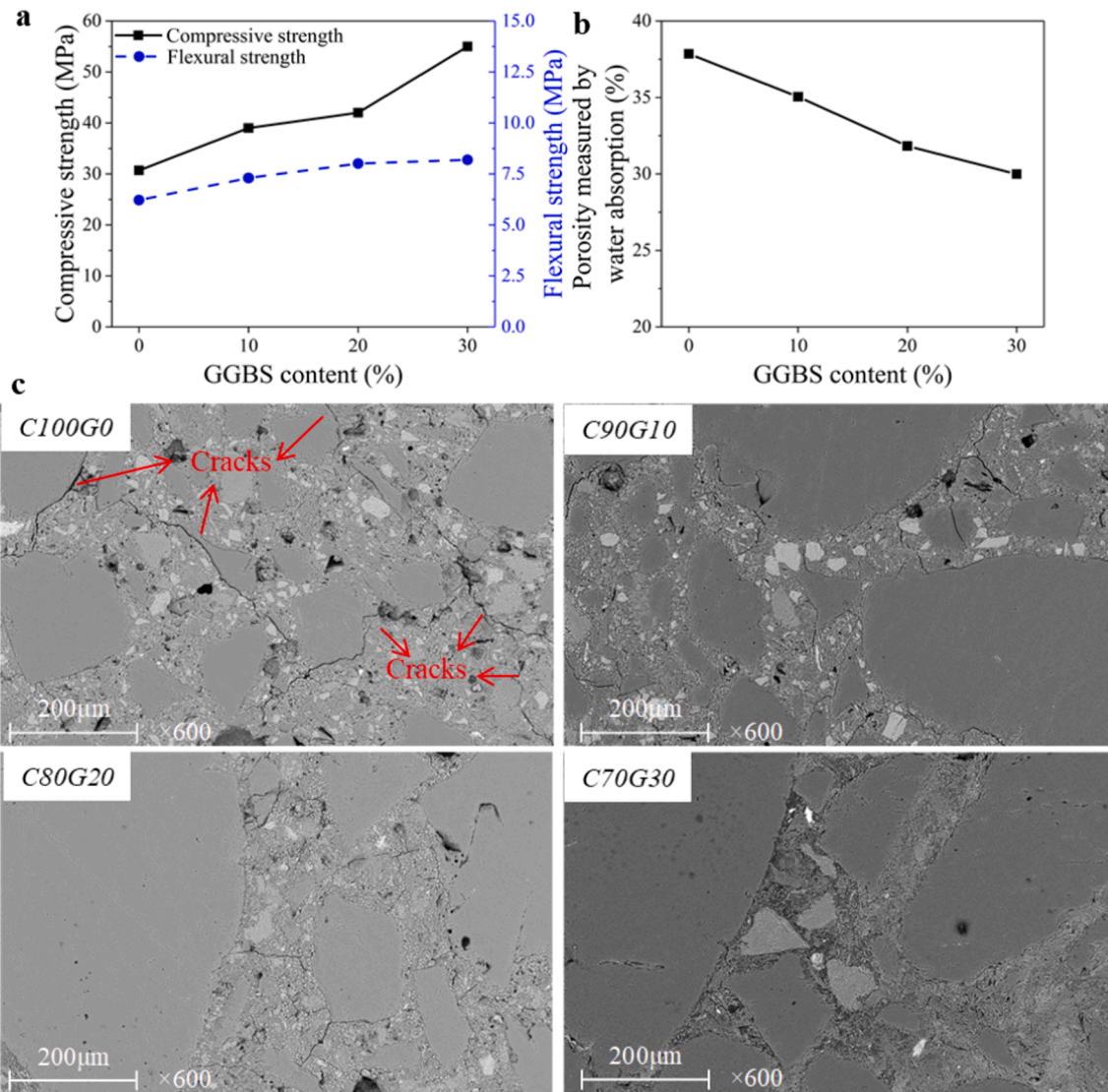


Fig. 1. (a) 28-day strength of alkali-activated calcined soil-GGBS mortar, (b) porosity of alkali-activated calcined soil-GGBS paste, and (c) SEM images of activated calcined soil-GGBS mortar with varying GGBS contents.

increased with higher GGBS contents. Compared to the alkali-activated pure calcined soil mortar sample, the compressive strength gradually increased by 21–79 %, and the flexural strength increased by 17–32 % as the GGBS content increased from 10 % to 30 %. The alkali-activated pure calcined soil mortar sample exhibited the lowest compressive and flexural strengths of 30.7 MPa and 6.2 MPa, respectively, while the mortar sample with 30 % of GGBS showed the highest compressive and flexural strengths of 55.0 MPa and 8.2 MPa, respectively. The increase in strength can be attributed to reduced porosity and finer pore sizes. As shown in Fig. 1b, the porosity of alkali-activated calcined soil-GGBS paste decreased from 37.9 % to 30.0 % with increasing GGBS content. Furthermore, SEM images in Fig. 1c reveal that as the GGBS content increased, the microstructure became denser. Fig. 2a shows cumulative porosity curves measured by MIP, and Fig. 2b shows the resulting pore size distribution. The porosity measured by MIP was slightly lower than that measured by the water absorption method, as gel pores absorb water and swell during the saturation process, altering the results. The MIP results also demonstrate a decrease in porosity with the addition of GGBS. Pores in cementitious products are classified as gel pores (less than 0.01 μm), medium capillary pores (0.01 to 0.5 μm), and large capillary pores (more than 0.5 μm) (Zhang et al., 2012). Fig. 2b indicates that gel porosity was significantly changed with the addition of GGBS,

possibly due to changes in the gel type. Large capillary pores significantly influence the strength and permeability of concrete (Igarashi et al., 2005). As shown in Fig. 2b, after adding GGBS, while total porosity decreased by 19.5 % from 34.9 % to 28.1 %, large capillary porosity decreased more substantially, from 25.8 % to 10.9 %, i.e. with a decrease of 57.8 %.

Fig. 2c and d show the efflorescence formation potential of alkali-activated calcined soil-GGBS paste with varying GGBS contents, indicating that the efflorescence resistance was enhanced after adding GGBS. Visually, only C100G0 showed whitish efflorescence products, as evident in Fig. 2c. And it is seen from Fig. 2d that the leached Na^+ concentration decreased with increasing GGBS contents. The cause of efflorescence formation is considered the presence of residual alkali in the pore solution of AAMs, as indicated by the results of the leaching tests. However, the presence of residual alkali was not caused by an excess of added alkali in this work. Based on the Na_2O to precursor ratio of 7 % and the reactive Al_2O_3 content of 23.1 % in the calcined soil, the initial Na/Al molar ratio in sample C100G0 was of about 0.5. This value is much lower than the theoretical upper limit value of $\text{Na}/\text{Al} = 1$ for AAMs, which accounts for Na^+ required to balance AlO_2^- sites. Despite the absence of an excess of added alkali, there is still much free alkali residue in the pore solution due to the incomplete reaction between

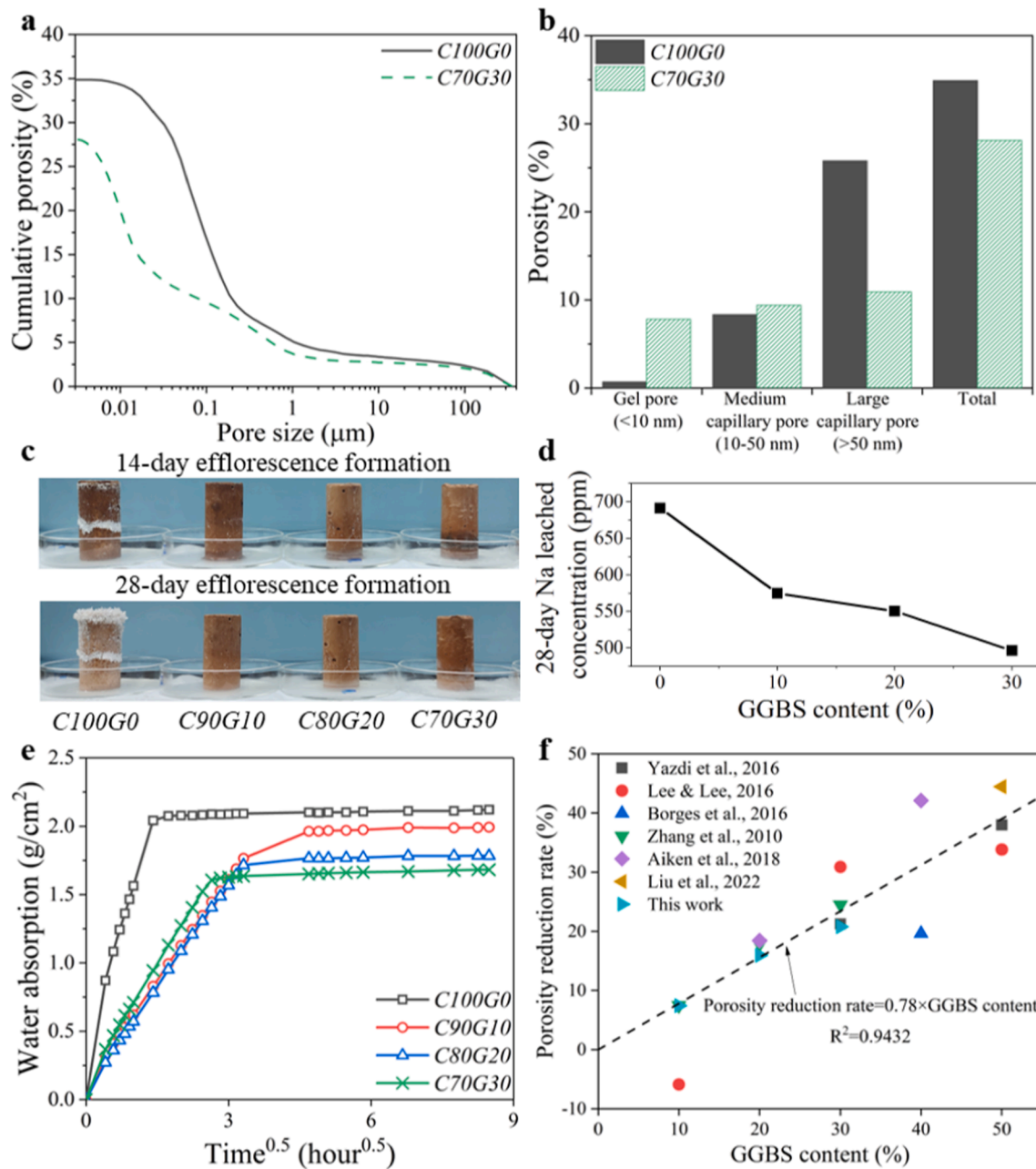


Fig. 2. (a) Cumulative porosity curves, (b) pore size distribution, (c) efflorescence formation, (d) efflorescence potential, (e) water absorption rate of alkali-activated calcined soil-GGBS paste with varying GGBS contents, and (f) relationship between AAMs' porosity and GGBS content reported in various studies.

precursor and alkali (Williamson and Juenger, 2016). In this work, the effective mitigation of efflorescence formation after GGBS addition can be attributed to the improved pore structure that hinders the movement of free Na^+ ions. As the GGBS content increased, the total porosity and large porosity decreased (Fig. 1b and Fig. 2b), and the microstructure became denser (Fig. 1c). Also, Fig. 2e shows that the ratio of water absorption to the square of time significantly decreased after GGBS addition. This reduction in absorption rate indicates a more tortuous pore structure.

A similar decreasing trend of the porosity of AAMs with increasing GGBS content has been observed in previous studies, as shown in Fig. 2f (Aiken et al., 2018; Borges et al., 2016; Lee and Lee, 2016; Liu et al., 2022; Yazdi et al., 2018; Zhang et al., 2010). The observed pattern reveals that porosity decreases by approximately 7.8 % with every 10 % increase in GGBS content. This reduced porosity is likely attributed to the coexistence of calcium aluminosilicate hydrate gel (C-A-S-H) and

sodium aluminosilicate hydrate gel (N-A-S-H). Specifically, C-A-S-H, with a significantly higher chemically bound water content than N-A-S-H, offers greater pore-filling capacity, leading to decreased porosity (Provis et al., 2012).

The alkali activation of calcined soil in this study follows the above mentioned mechanism, supported by the following empirical evidence shown in Fig. 3. Fig. 3a illustrates FTIR absorbance spectra of alkali-activated calcined soil-GGBS paste with varying GGBS contents. The chemical species change of AAMs can be identified by FTIR spectra. The alkali-activated products show a peak or peaks within the wavenumbers $900 \sim 1100 \text{ cm}^{-1}$ because of the asymmetric stretching vibration in the Si-O-Al structure ($\nu_{\text{as}}\text{Si-O-Al}$). The peak wavenumber of C-A-S-H is lower than that of N-A-S-H (Garcia-Lodeiro et al., 2011); therefore, the peak at 1008.6 cm^{-1} is attributed to C-A-S-H, while N-A-S-H corresponds to the peak at 1031.7 cm^{-1} . As the GGBS content increased, Fig. 3a demonstrates a gradual increase in the relative peak height of C-A-S-H to N-A-S-

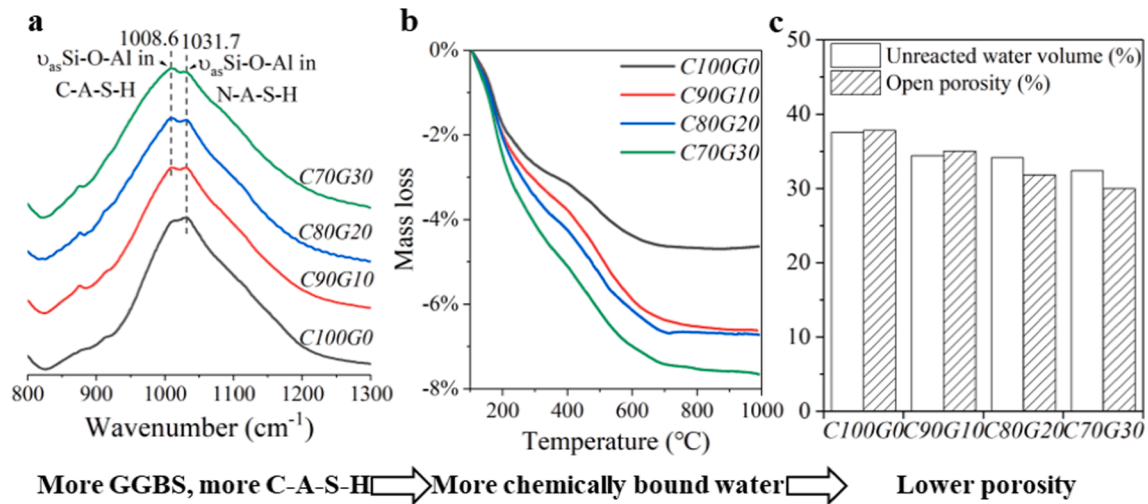


Fig. 3. (a) FTIR absorbance spectra, (b) thermogravimetric curves, and (c) unreacted water volumes and open porosity of alkali-activated calcined soil-GGBS paste with varying GGBS contents.

H. The peak height is directly proportional to the content of the respective species. Thus, as the GGBS content increased, the C-A-S-H content increased. Fig. 3b displays the thermogravimetric curves, indicating an increase in chemically bound water content (mass loss from 105 °C to 600 °C) with a higher GGBS content. As shown in Fig. 3b, the weight loss was observed up to 600 °C for sample C100G0, whereas it extended to 700 °C after the addition of GGBS. The gel loses chemically bound water below 600 °C (Burciaga-Díaz and Escalante-García, 2017; Chen et al., 2024; Rivera et al., 2016). Weight loss between 600 °C and 700 °C is likely due to the decomposition of calcium carbonate in GGBS (Burciaga-Díaz and Escalante-García, 2017). The unreacted water forms pores and reduces strength (Zhou et al., 2024b). Therefore, an increased chemically bound water content indicates a decreased unreacted water volume and, consequently, a lower porosity. Fig. 3c compares open porosity with unreacted water volume, revealing their close correlation. The unreacted water volume in Fig. 3c was calculated as the initially added water content minus the chemically bound water content, as shown in Table S3 in the Supplementary Materials. In summary, increased GGBS content results in more C-A-S-H, and more chemically bound water in C-A-S-H compared to N-A-S-H leads to reduced porosity and refined pore size, ultimately enhancing both strength and efflorescence resistance. The microcracks at the paste/particle interface in Fig. 1c were likely caused by the shrinkage of AAMs. This shrinkage may result from the absolute volume reduction during the alkali-activation reaction (Li et al., 2019b). Compared to AAMs prepared with blends of calcined soil and GGBS, those made with 100 % calcined soil contained less chemically bound water, resulting in a lower volume of reaction product and thus leading to more pronounced shrinkage and noticeable microcracks at the paste/particle interface.

Although adding GGBS or other Ca sources is feasible in mitigating efflorescence, two conditions must be met: 1) the GGBS content should be no less than 10 % to provide sufficient Ca; 2) the GGBS content should not be excessive to ensure that the Na/Al ratio remains below 1. Consistent with this study, Zhang et al. (2024) and Zhang et al. (2014) reported effective efflorescence mitigation in alkali-activated fly ash with GGBS contents of 20 %–40 %. However, lower GGBS contents were found to be ineffective. Longhi et al. (2021) observed that a 5 %–10 % replacement of metakaolin with low-CaO GGBS did not effectively mitigate the efflorescence of alkali-activated metakaolin. This lack of improvement can be explained by insufficient Ca. In addition, introducing excessive Ca can also be ineffective. Matakah et al. (2021) reported that a 10 % replacement of metakaolin with Portland cement reduced the efflorescence of alkali-activated metakaolin, whereas a 15 % replacement was ineffective. This lack of improvement may be

attributed to insufficient Al. In their study, the Na/Al molar ratios of the alkali-activated metakaolin with cement contents of 10 % and 15 % were 0.97 and 1.03, respectively. The 15 % replacement increased the Na/Al ratio above 1, exceeding the theoretical upper limit for AAMs, resulting in significant alkali residue and subsequent efflorescence. Srinivasamurthy et al. (2022) and Santana et al. (2023) have emphasized that an excessive Na/Al ratio significantly promotes efflorescence formation. Similar to cement, GGBS also contains much less Al than calcined clay. Increasing GGBS content increases the Na/Al ratio; therefore, excessive GGBS addition may lead to efflorescence formation.

The influence of GGBS content on the early performances (fluidity, setting time, and one-day strength) is shown in Fig. S4. From Fig. S4a, it can be seen that the flow diameter increased from 160 mm to 198 mm as the GGBS content increased from 0 to 30 %. The enhanced fluidity can be attributed to the reduced calcined soil content. Calcined soil is rich in metakaolin. Previous studies have reported that metakaolin, with its high specific surface area and plate-like shape, requires more water to form the water layer on the particle surface, thus leading to poor workability of both cement-based materials containing metakaolin and alkali-activated metakaolin (Muduli and Mukharjee, 2019; Provis et al., 2010; Zhang et al., 2022). In Fig. S4b, a substantial decrease in setting time, from 412 min to 20–40 min after GGBS addition, can be observed. Li et al. (2017, 2019a) attributed this phenomenon to the high CaO content of GGBS. The low energy of Ca–O and Al–O bonds enables the rapid release of Ca and Al, leading to the formation of C-A-H and accelerating the setting process. This explanation also accounts for the high early strength of samples with GGBS, as depicted in Fig. S4c. Increasing the GGBS content from 0 to 30 % significantly enhanced the one-day compressive strength from 13.6 MPa to 38.1 MPa and the one-day flexural strength from 2.81 MPa to 5.99 MPa. The remarkable early strength of sample C70G30 makes it particularly suitable for pavement construction and repair, facilitating shortened construction times and swift traffic resumption.

3.2. Influence of the uncalcined soil content on the performance of the concrete samples

Fig. 4a shows the compressive strength variations of alkali-activated calcined soil-uncalcined soil-GGBS concrete with the uncalcined soil content. As the uncalcined soil content increased, the one-day and 28-day compressive strength decreased. However, notably, sample C52.5G30U17.5 with 17.5 % uncalcined soil content exhibited only a slight reduction in compressive strength compared to sample C70G30U0. It still achieved a compressive strength of 36 MPa after one

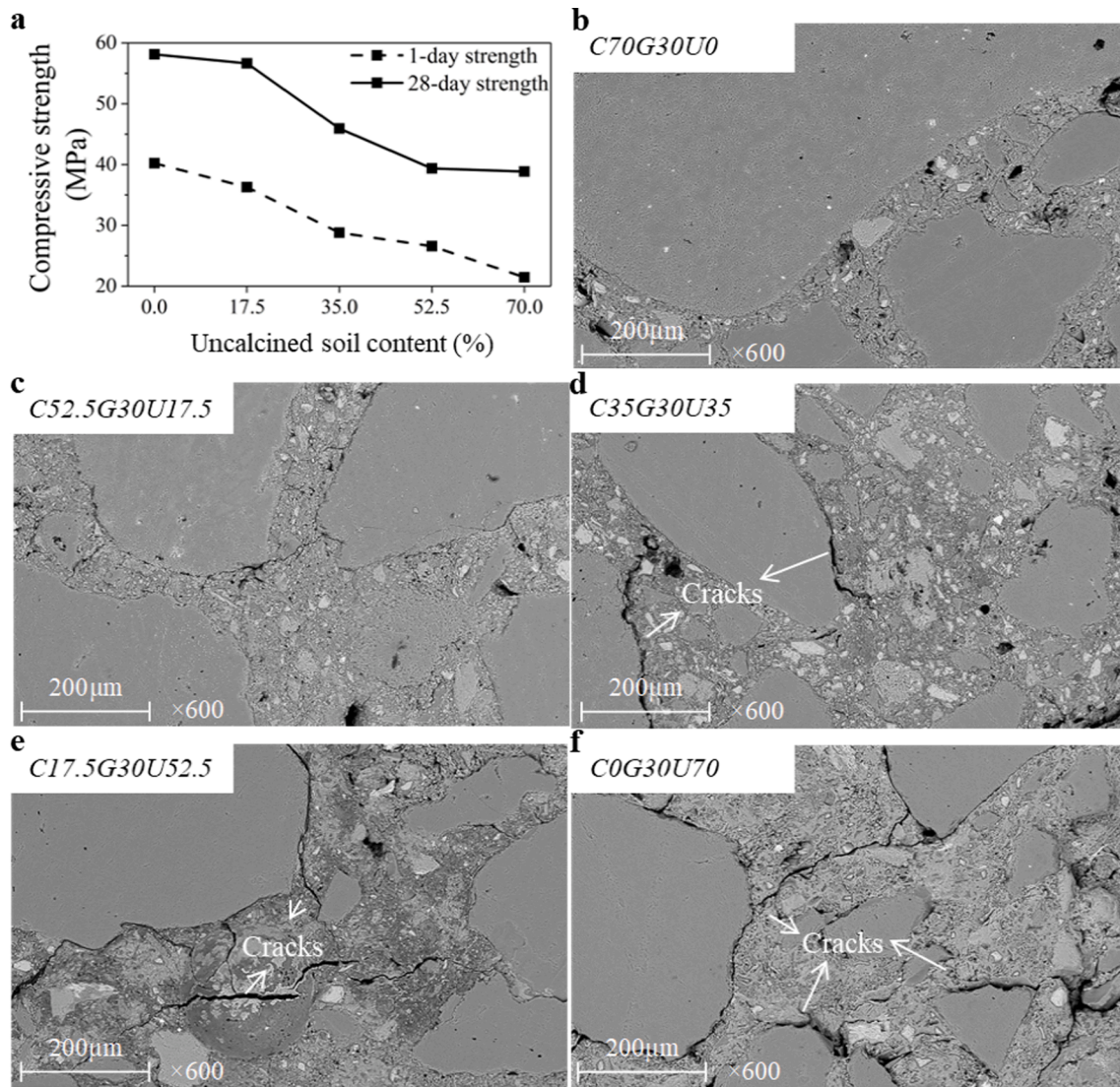


Fig. 4. (a) Compressive strength and (b-f) SEM images of alkali-activated calcined soil-uncalcined soil-GGBS concrete with varying uncalcined soil contents.

day and 56 MPa after 28 days, showing excellent potential for use in pavement concrete. Only when the uncalcined soil content exceeded 35 %, the strength significantly decreased. SEM images in Fig. 4(b-f) reveal that the samples C70G30U0 and C52.5G30U17.5 (uncalcined soil content ≤ 17.5 %) exhibited denser microstructures, while many visible cracks appeared in samples C35G30U35, C17.5G30U52.5, and C0G30U70 (uncalcined soil content ≥ 35 %), which is consistent with the strength variations.

Fig. 5a presents the FTIR absorbance spectra of the uncalcined soil, paste sample C70G30U0, and paste sample C52.5G30U17.5. Apart from the peaks around 1000 cm^{-1} due to the asymmetric stretching vibration in the Si-O-Al bond ($\nu_{\text{as}}\text{Si-O-Al}$), uncalcined soil had two distinctive peaks: 912.2 cm^{-1} corresponding to the bending vibration of the Al-OH bond ($\delta\text{Al-OH}$), and 1114.7 cm^{-1} corresponding to the asymmetric stretching vibration of the Si-O-Si bond ($\nu_{\text{as}}\text{Si-O-Si}$). These two peaks were also observed in C52.5G30U17.5. Peak intensity in an infrared absorbance spectrum is directly proportional to the substance's content. In C52.5G30U17.5, the initial uncalcined soil content was 17.5 %. By subtracting 17.5 % of the uncalcined soil absorbance spectrum from that of C52.5G30U17.5, the two peaks just disappeared, indicating the complete inertness of uncalcined soil. Therefore, replacing calcined soil with uncalcined soil introduced many inert particles, while most cracks occurred at the interfaces between the particles and the matrix, as shown

in the SEM images in Fig. 4. The cracks not only reduce strength, but also provide channels for the water and ion movement, increasing water absorption and reducing durability. However, it is noted in Fig. 5a that the height of the C-A-S-H peak in C52.5G30U17.5, after the subtraction of the uncalcined soil spectrum, was higher compared to C70G30U0, while the height of the N-A-S-H peak decreased. This result suggests an increment in the C-A-S-H content and a decrease in the N-A-S-H content, leading to the higher chemically bound water content and reduced porosity resulting for C52.5G30U17.5. The uncalcined soil is chemically inert, so the increased C-A-S-H content can be explained by the increased proportion of GGBS in the calcined soil-GGBS blends when replacing calcined soil with uncalcined soil. Despite its inertness, uncalcined soil has two important roles: it acts as a filler and reduces the use of calcined soil, which in turn reduces the costs and carbon emissions associated with calcination; additionally, because calcined soil's metakaolin content, with its high specific surface area and plate-like shape, hampers fluidity, replacing calcined soil with uncalcined soil improves fluidity, as shown in Fig. S5 in the Supplementary Materials.

The thermogravimetric curves in Fig. 5b confirm the increased chemically bound water content in C52.5G30U17.5. In addition, the MIP and water absorption results in Fig. 5(c-e) show the slightly reduced porosity and similar pore size of C52.5G30U17.5 compared to C70G30U0. When replacing calcined soil with a small amount of

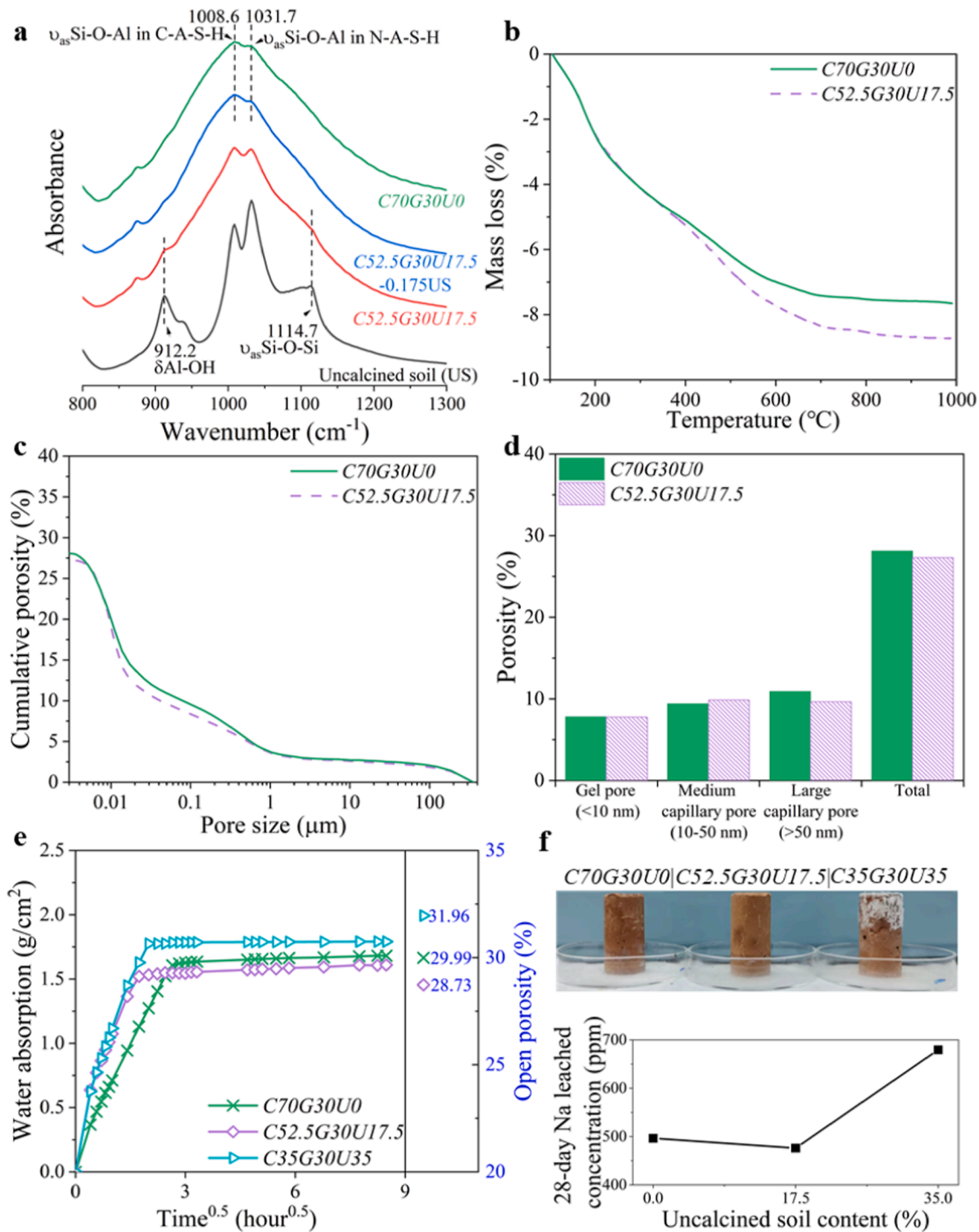


Fig. 5. (a) FTIR absorbance spectra of the uncalcined soil and paste samples, (b) thermogravimetric curves, (c) cumulative porosity curves, (d) pore size distribution, (e) water absorption and open porosity, and (f) 28-day efflorescence formation and potential of paste samples.

uncalcined soil in C52.5G30U17.5, the decreased porosity caused by the increased C-A-S-H content compensates for the increased porosity because of the cracks in the interfaces between the uncalcined soil and the matrix, preventing strength loss, and even enhancing efflorescence resistance. Fig. 5f demonstrates that neither C70G30U0 nor C52.5G30U17.5 exhibited efflorescence formation, with C52.5G30U17.5 displaying lower efflorescence potential. However, when the uncalcined soil content exceeded 35 %, the decreased porosity due to the increased C-A-S-H content cannot compensate for the increased porosity due to the cracks in the interfaces between the

uncalcined soil and the matrix, reducing strength and increasing porosity. C35G30U35 exhibited high porosity (Fig. 5e), and its decreased calcined soil content led to low reactive Al content and thus high free Na⁺ content, resulting in low efflorescence resistance.

In summary, a low uncalcined soil replacement rate of 25 % (17.5 % uncalcined soil content and 52.5 % calcined soil content) is recommended. This replacement rate is comparatively lower than that reported for other types of waste such as limestone powder, basalt powder, waste marble powder, and recycled aggregate powder (Kabirova et al., 2023; Liu et al., 2023; Miyan et al., 2024; Omur et al., 2024; Perez-

Cortes and Escalante-Garcia, 2020). This discrepancy arises from the inert nature of the uncalcined soil used in this study, in contrast to limestone powder, basalt powder, waste marble powder, and recycled aggregate powder, which can engage in the alkali-activation process to form AAMs (Mao et al., 2022; Ortega-Zavala et al., 2019; Sasui et al., 2023). The inertness of uncalcined soil restricts further substitution, whereas calcination, though effective in activating soil, consumes energy and leads to increased CO₂ emissions. Future research should explore low-cost and low-carbon methods for activating soil waste. In the field of alkali-activated clays, technologies such as mechanical activation have already been explored (Mañosa et al., 2024; Marsh et al., 2024; Yamchelou et al., 2021).

3.3. Carbon emissions and cost analysis

Fig. 6 shows the CO₂ emissions and costs of the developed concrete based on these alkali-activated soil-GGBS blends in comparison with conventional cement concrete presenting a similar strength grade. The strength of the concrete produced in this study reached a maximum of 58 MPa, so the reference cement concrete was assumed to present a 55 MPa strength, with a water/binder ratio of 0.33, using 30 % fly ash and 70 % cement as the binder. The detailed mixture proportions of these concrete products are given in Tables S4 and S5 in the Supplementary Materials.

From Fig. 6, it can be seen that the CO₂ emissions of the reference cement concrete were 314.9 kgCO₂/m³, while the CO₂ emissions of the developed concrete in this study were only 143.3–211.7 kgCO₂/m³ with similar costs. Because the price of GGBS is higher than that of calcined soil, the cost of the developed concrete increased with the GGBS content. However, due to the improved strength with the addition of GGBS, the cost per MPa significantly decreased as the GGBS content increased. Additionally, GGBS has a lower CO₂ emission factor compared to calcined soil, resulting in a significant reduction in CO₂ emissions per MPa as the GGBS content increased. While uncalcined soil has a lower price and a lower CO₂ emission factor compared to calcined soil, adding a significant amount of uncalcined soil reduced the performance of the developed concrete, leading to no reduction in CO₂ emissions per MPa and even a higher cost per MPa. However, it is important to note that C52.5G30U17.5 exhibited only a slight decrease in compressive strength compared to C70G30, resulting in a comparable cost per MPa and lower CO₂ emissions per MPa.

Among the tested concrete formulations, C70G30 (calcined soil: GGBS:uncalcined soil = 70:30:0) and C52.5G30U17.5 (calcined soil: GGBS:uncalcined soil = 52.5:30:17.5) exhibited the lowest CO₂ emissions per MPa and cost per MPa. They achieved a 42 % reduction in CO₂ emissions and similar costs compared to the reference concrete considered. The preliminary sustainability assessment shows the potential of alkali-activated soil-GGBS blends as sustainable construction materials.

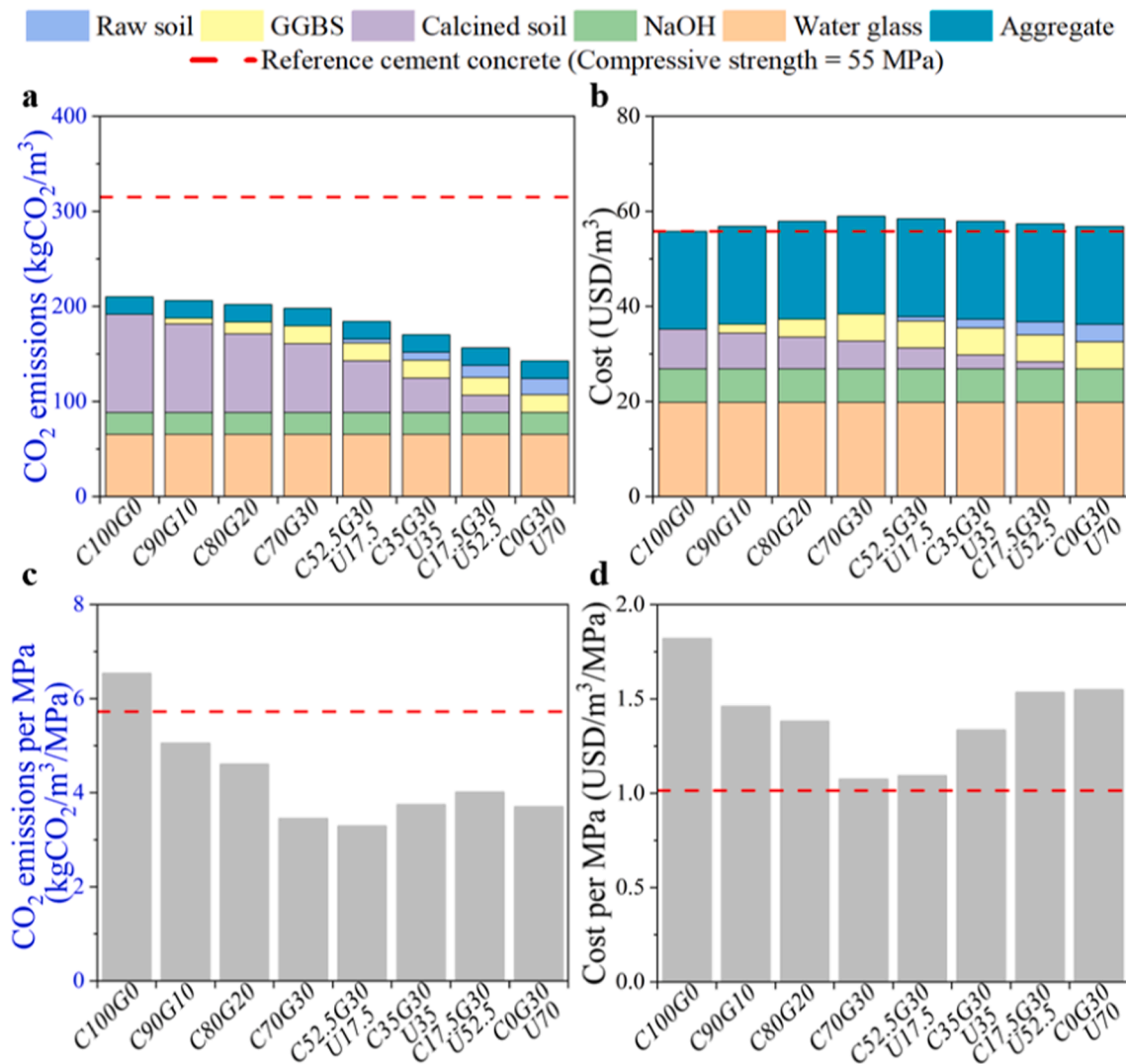


Fig. 6. CO₂ emissions and costs of the developed concrete based on alkali-activated soil-GGBS blends.

3.4. Multi-criteria performance evaluation

The developed alkali-activated soil-GGBS blends were evaluated using radar charts, assessing five main factors: 28-day compressive strength, efflorescence potential, open porosity, cost per MPa, and CO₂ emissions per MPa, as depicted in Fig. 7a. Here, the five indicators were normalized: the normalized compressive strength of a sample was the ratio of its strength to the maximum strength among samples; for the other four factors, the normalization result of a sample was the ratio of the minimum value among all samples to the sample value. From Fig. 7a, it can be seen that C52.5G30U17.5 had the largest area (1.88) and was the optimal option. This formulation was compared with the soil waste-based construction materials reported in other studies (Hu et al., 2022; Santos Nascimento et al., 2021; Shen et al., 2024; Wu et al., 2022; Zhou et al., 2021, 2023a), as shown in Fig. 7b. Here, the functional unit was one ton of binder or one ton of brick. Among these soil waste-based

construction materials, the material developed in this study, C52.5G30U17.5, had the highest compressive strength and the lowest CO₂ emissions per MPa. These advantages show that the method investigated in this study appears to be a promising option for recycling soil waste in construction materials with low CO₂ emissions.

4. Conclusions

This study presented a novel pathway toward large-scale recycling of excavated soil waste. The process involved activating soil waste through calcination to convert kaolin to metakaolin, subsequently using it to produce AAMs, supplemented with GGBS and uncalcined soil waste for enhanced performance and sustainability. Key findings include:

- 1) The AAMs prepared with 100 % calcined waste soil showed middle strength and a comparatively high efflorescence risk. Partly

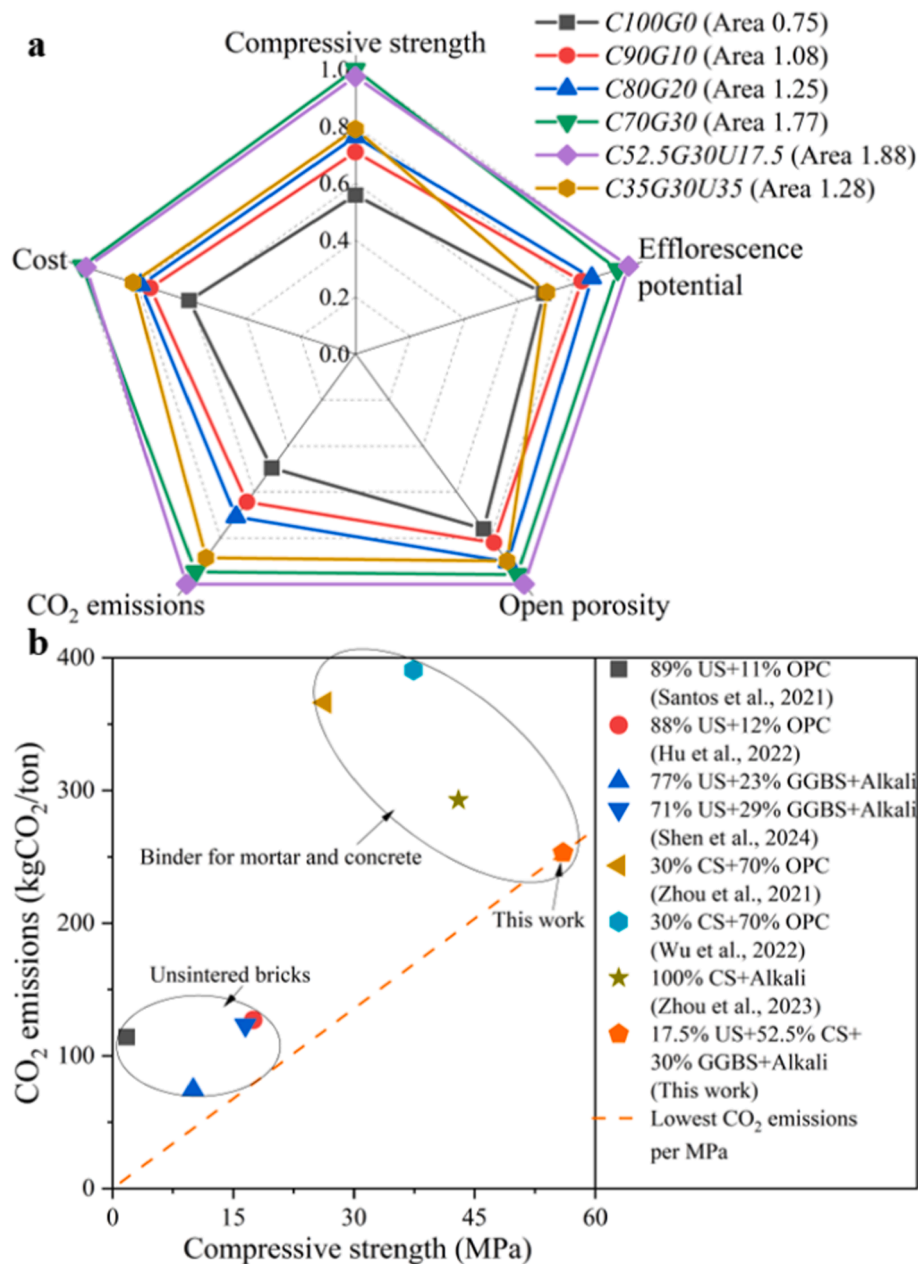


Fig. 7. (a) Multi-criteria performance evaluation of alkali-activated soil-GGBS blends and (b) comparison of various paths of recycling soil waste into construction materials (US: uncalcined soil waste; CS: calcined soil waste; GGBS: ground granulated blast furnace slag; OPC: ordinary Portland cement).

replacing calcined soil with GGBS improved the performance of AAMs and reduced the cost and CO₂ emissions per MPa. The main component was altered from N-A-S-H to a coexistence of N-A-S-H and C-A-S-H after adding GGBS. The C-A-S-H generation contributes to lower porosity by chemically binding more water, thereby enhancing strength and efflorescence resistance. However, a GGBS content exceeding 30 % risks flash setting, which should be carefully controlled. Also, GGBS addition should ensure that the Na/Al ratio does not exceed 1 to avoid efflorescence caused by an excess of added alkali.

- 2) When GGBS was added, replacing 25 % of the calcined soil waste with uncalcined soil waste can ensure no significant strength loss while enhancing efflorescence resistance and reducing CO₂ emissions per MPa. At this content, although uncalcined soil is an inert material that increases the weak interfaces in the matrix and weakens the mechanical properties, the increase in GGBS proportion in the calcined soil-GGBS blend increased the C-A-S-H content and thus reduced porosity, offsetting the negative effect. The inertness of uncalcined soil restricts further substitution. Future research should explore low-cost and low-carbon methods for activating soil waste, such as mechanical activation.
- 3) Considering the compressive strength, efflorescence performance, porosity, cost per MPa, and CO₂ emissions per MPa, the alkali-activated materials containing 52.5 % calcined soil, 30 % slag, and 17.5 % uncalcined soil exhibited the best overall performance. It also showed proper fluidity and outperformed the soil waste-based construction materials reported in other studies in terms of compressive strength and CO₂ emissions per MPa. Compared to the reference blended cement concrete, the developed concrete based on the alkali-activated calcined soil-uncalcined soil-GGBS blends exhibited high early strength, 42 % lower CO₂ emissions, and a similar cost. These findings suggest that this recycling method appears to be a promising option for recycling excavated soil waste and reducing reliance on carbon-intensive cement in the construction sector. It should be noted that the sustainability evaluation performed in this study was preliminary. A complete life cycle assessment should be performed in future studies, as well as a more detailed life cycle costing evaluation.

A key issue for large-scale recycling is that soils from different regions may differ in chemical and mineral compositions and thus result in variations in the final AAMs despite using the same mixture proportions (including alkali-activated solution to soil ratio, GGBS to soil ratio, as well as H₂O/Na₂O and SiO₂/Na₂O ratios in the alkali-activated solution) with this study. However, the reactive oxide composition can serve as a useful reference for adapting mixture designs to different soils. Formulating the mixture proportions based on the reactive oxide composition is expected to produce samples with similar performances using different soils. From a microscopic perspective, in the synthesis of AAMs, calcined soil primarily contributes reactive SiO₂ and Al₂O₃, the alkali-activator supplies Na₂O and supplements SiO₂, GGBS provides reactive CaO and additional SiO₂ and Al₂O₃, and water serves as the reaction medium while making the mixture flowable. Based on the reactive components of the materials used in this study, the optimal mix corresponds to a CaO/SiO₂ molar ratio of 0.52, a SiO₂/Al₂O₃ molar ratio of 3.2, a Na₂O/Al₂O₃ molar ratio of 0.71, and a H₂O/(CaO + Na₂O + SiO₂ + Al₂O₃) mass ratio of 0.44. Considering these oxide ratios and the reactive components of the materials in other scenarios, the corresponding mixture proportions can be calculated. The uncalcined soil to calcined soil ratio can be set to 1:4, because uncalcined soil does not participate in alkali-activation and primarily contributes through a filling effect. To ensure uniformity, pre-homogenization of soil is essential for large-scale recycling. Trial mixing is necessary to validate the calculated mixtures. Further research into the influence of soil composition on the performance of AAMs is necessary to guide soil recycling across diverse regions. Mixing soils from different regions in

appropriate proportions may optimize the reactive oxide composition of the soil and thus reduce GGBS and alkali-activator dosage.

CRedit authorship contribution statement

Chenchen Luan: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Ao Zhou:** Writing – review & editing, Visualization, Supervision, Project administration, Conceptualization. **Tiejun Liu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Dujian Zou:** Writing – review & editing. **Pan Gao:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2025.02.030>.

Data availability

Data will be made available on request.

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